

Homework 3: Bifurcations and Hysteresis

MATH 515 — Fall 2026

Due Friday, October 2, 2026

Assignment convention. Each numbered problem reproduces the full assigned statement from Steven H. Strogatz, *Nonlinear Dynamics and Chaos*, third edition (the specific section/problem number is given parenthetically), with some mild editing for clarity. Any additional requirements are labeled **Course-specific instructions** and are part of the assignment.

Problem 1: Climate tipping

Textbook problem (Strogatz, third edition, Exercise 3.1.6, “Climate model revisited”):

Consider the following energy-balance model for the Earth’s mean surface temperature T :

$$C \frac{dT}{dt} = (1 - \alpha)Q - \varepsilon \sigma T^4.$$

For simplicity, all the parameters are sometimes assumed independent of T . The parameter α (known to climate scientists as the “planetary albedo”) quantifies how reflective the Earth’s surface is; α is defined as the fraction of the incoming solar radiation that gets reflected, on average. Thus $0 < \alpha < 1$.

But as many of us have learned in this era of concerns about climate change, ice is much more reflective than land or open water, which raises the specter of a runaway ice-albedo feedback loop: the warmer the climate gets, the more the glaciers and polar ice caps will tend to melt, which means the Earth will become less reflective, so even more solar radiation will get absorbed and the Earth will get even hotter. Conversely, the colder the planet gets, the more ice can form, increasing the reflectivity of the Earth and cooling it even more.

Building on the work of Budyko (1969) and Sellers (1969), Fraedrich (1978) introduced the simple model above and incorporated an ice-albedo feedback in it by assuming that α decreases with rising temperature:

$$\alpha = a - bT,$$

where $a > 1$ and $b > 0$ are constants. Thus the model becomes

$$C \frac{dT}{dt} = (1 - [a - bT])Q - \varepsilon \sigma T^4.$$

- Analyze the vector field for T . Show graphically that the model can have two fixed points, or one, or none, depending on the value of Q .
- How is $T(t)$ predicted to behave in all three cases? Be sure to comment on how your results depend on where the initial temperature $T(0)$ lies in relation to the fixed points.
- Show that a saddle-node bifurcation occurs at a critical value $Q = Q_c$ (which you should calculate).

For more about abrupt changes in climate associated with bifurcations and tipping points, including data from eight ancient climate shifts, see Dakos et al. (2008).

Course-specific instructions. Complete all parts on the physical domain $T \geq 0$. Assume $C, \varepsilon, \sigma > 0$ and $Q \geq 0$, in addition to the stated assumptions $a > 1$ and $b > 0$. Use the linear albedo approximation $\alpha = a - bT$ on all $T \geq 0$, even where its values fall outside the physically meaningful interval $(0, 1)$. If the model predicts cooling below $T = 0$, interpret this as a breakdown of the model.

Problem 2: Hysteresis

Textbook problem (Strogatz, third edition, Exercise 3.4.14, “Subcritical pitchfork”):

Exercise 3.4.14 (Subcritical pitchfork). Consider the system

$$\dot{x} = rx + x^3 - x^5,$$

which exhibits a subcritical pitchfork bifurcation.

- Find algebraic expressions for all the fixed points as r varies.
- Sketch the vector fields as r varies. Be sure to indicate all the fixed points and their stability.
- Calculate r_s , the parameter value at which the nonzero fixed points are born in a saddle-node bifurcation.

Course-specific instructions. Complete all parts. In part (c), which has confusing wording as it appears in the textbook, instead use your answers to (a) and (b) to determine the smallest value of r for which nonzero equilibria exist. Find those equilibria and describe how the nearby equilibria and their stability change as r increases through this value, identifying the bifurcation type.

Problem 3: A two-parameter imperfect pitchfork

Textbook problem (Strogatz, third edition, Exercise 3.6.3, “A perturbation to the supercritical pitchfork”):

Consider the system

$$\dot{x} = rx + ax^2 - x^3,$$

where $-\infty < a < \infty$. When $a = 0$, we have the normal form for the supercritical pitchfork. The goal of this exercise is to study the effects of the new parameter a .

Course-specific instructions. Here, you may note that the actual directions from the textbook problem are omitted. This is because the problem, as it is originally stated, is somewhat excessively long to do (but also pretty cool). We will do a smaller version here. Assume $a = 1$ and do the following:

- Find all equilibria and determine their stability as r varies.
- Sketch the bifurcation diagram, distinguishing stable and unstable branches.
- Find and classify every bifurcation.

(Note: The problem, as originally stated in the text, has you do this for a generic value of the parameter a and then asks you to determine the values at which your answer changes qualitatively.)